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A quantum group decision model for meteorological disaster emergency response based on D-S evidence theory and Choquet integral

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ABSTRACT

In addressing complex and dynamic meteorological disaster decision-making environment, the traditional multi-attribute group decision-making domain model is often unable to effectively deal with the correlation between attributes and the mutual influence of group opinions. To overcome this challenge, this paper proposes a novel quantum framework for group decision-making, which is used to deal with the emergency situation of meteorological disaster. The model initially characterizes attribute correlations using 2-additive Choquet integrals and employs Dempster-Shafer evidence theory to both integrate information and ascertain attribute weights, and decision makers' weights are calculated based on grey relative correlation. On this basis, a quantum-like Bayesian network is developed to capture the interference among decision-makers' opinions. The alternatives are ranked by quantum probabilities computed based on Bayesian principle. Finally, a case study on meteorological disaster emergency scenario assessment is conducted to validate the proposed model's effectiveness and superiority. Additionally, its stability and practicality are confirmed through sensitivity analysis and comparative analysis.

1. Introduction

The problem of meteorological disaster emergency decision-making requires the participation of multiple agents to collect multi-directional information of disaster scenes, falling within the domain of multi-attribute group decision-making. For instance, when dealing with natural disasters like floods or typhoons, establishing an efficient multi-attribute group decision-making mechanism is vital for the timely and effective execution of rescue operations [1–3]. In practical scenarios, leaders frequently convene teams of experts to address multi-attribute decision-making challenges, necessitating a delicate trade-off among various evaluative factors to identify the optimal solution [4]. The decision-making process begins with aggregating diverse assessments of alternatives by team members, facilitating a comprehensive, multidimensional analysis [5].

In multi-attribute decision-making, ascertaining attribute weights is a pivotal stage to ensure decision accuracy and reliability. Since attributes can exert varying influences on the ultimate decision, allocating suitable weights to them is fundamental for effec-

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tive decision-making [6]. Although Analytic Hierarchy Process, entropy weight method, TOPSIS method and other methods have been widely used [7–9], they show some limitations in dealing with the correlation between attributes and eliminating the subjective preferences among members of the decision group. While, Choquet integral method shows obvious advantages in the field of multi-attribute decision making because of its comprehensive consideration of the interaction between attributes. This methodology can proficiently model and analyze decision makers' preferences by incorporating the Choquet integral into fuzzy measurements [10]. For instance, the Choquet integral measure is used to address multi-attribute decision-making issues involving multiple information sources [11,12]. In the context of probabilistic hesitant fuzzy environments, the TODIM method, rooted in the Choquet integral, is formulated. The Choquet integral offer fresh resources for decision-making, information integration, game theory, data mining, and various other domains [13]. The significance and validity of the Choquet integral in multi-attribute decision-making are substantiated, so it is widely used in the field of multi-attribute decision making [14,15].

In group decision-making problems, the opinions of decision makers affect and interfere with each other, and are often not independent. Traditional group decision-making methods typically rely on information aggregation or treat the decision-making process as a Bayesian network, grounded in conventional probability theory [16]. Conversely, quantum probability theory, as an extension of traditional probability theory, offers enhanced precision in modeling the interference among individual perspectives within complex systems [17–19]. This framework proficiently models paradoxical, suboptimal choices, and irrational human decision-making behaviors [20], primarily influencing probability calculations through quantum interference effects [21], As a result, decision analysis within the quantum theory framework presents substantial advantages. For instance, a decision-making framework based on quantum spherical fuzzy environments enhances decision quality by conducting in-depth analyses of the perception process [22,23]. In the realm of prediction, researchers have endeavored to integrate prediction into quantum frameworks, aiming to enhance prediction speed and accuracy [24]. Additionally, some studies have demonstrated the computational advantages of quantum computing over traditional methods in Bayesian network analysis [25]. Consequently, this method not only improves the decision quality, but also ensures the comprehensiveness and accuracy of the results.

In the decision-making process, due to the influence of the expert's personal preference and decision focus, there are often significant differences between the decision results. Dempster-Shafer evidence theory (D-S evidence theory) can amalgamate decision-making experts' perspectives, adeptly resolving evidence conflicts and conducting fusion [26]. This theory exhibits notable strengths in addressing uncertainty and fuzzy problems [27], it has been widely used in fault diagnosis, risk assessment and other multi-attribute decision making fields [28–30]. For instance, the amalgamation of fuzzy D-S evidence theory and fuzzy rough set models enhances decision-making accuracy and noise tolerance [31]. Likewise, the combination of conditional information entropy and D-S evidence theory effectively accomplishes information fusion and attribute dimensionality reduction for incomplete mixed data from various sources [32]. Some scholars have explored complex D-S evidence theory via quantum representation and quantum belief entropy, introducing a novel approach for measuring uncertainty in multi-attribute decision-making [33]. These investigations introduce novel methodologies and tools for addressing uncertainty within intricate domains, offering valuable insights and techniques for both scientific research and practical applications.

The identification of motivations is pivotal to the study [34–36], and two main motivations of this article are listed as follows.

Motivation 1: To address the intricacies of attribute interactions in decision-making processes, this paper integrates the 2-additive Choquet integral with evidence theory. This integration aims to improve the objectivity in determining attribute weights and to more accurately capture the interdependencies among attributes, thereby enhancing the reliability and precision of the decision-making process.

Motivation 2: In order to reflect the interference between decision makers in multi-attribute group decision making problems, this paper presents a quantum framework model aggregate information to reflect the influence of mutual interference among decision makers on alternative selection.

Considering the motivations, we propose a quantum group decision model based on D-S evidence theory and Choquet integral, to carry out more accurate and comprehensive analysis in the complex decision-making environment, thus playing an important role in the field of group decision-making.

The main contributions of this paper are as follows.

1) A multi-attribute quantum group decision model is constructed, which can reflect the mutual interference effect between the decision-makers' views, so as to be more in line with the complexity of the actual decision process.
2) A 2-additive Choquet integral based on Mobius transformation is used to deal with the interactions between attributes. Based on this, a new method is proposed to determine the attribute weights used to characterize these interactions.
3) Grey correlation analysis is used to reveal the correlation between the evaluation of different decision makers and the overall evaluation information, and then determine the weight of decision makers, and provide the probability information of key nodes for the quantum Bayesian framework.
4) In order to solve the problem of evidence conflict, D-S evidence theory is used to integrate different information sources and convert these integrated information into conditional probability in quantum Bayesian framework, which provides an important basis for determining quantum probability.

This paper is structured as follows: section 2 provides a review of 2-additive Choquet integrals, D-S evidence theory, and the quantum probabilistic framework. Section 3 introduces the Multi-Attribute Quantum Group Decision Model (MAQGDM), which combines the 2-additive Choquet integral, D-S evidence theory, quantum probability, and Bayesian network. In section 4, the steps of MAQGDM and its comparison with other existing decision models are introduced. In section 5, the MAQGDM are illustrated by emergency rescue cases of meteorological disasters, and the stability and reliability of the model are verified by sensitivity analysis and comparative analysis. Finally, section 6 summarizes the main findings and contributions of the research, and puts forward the prospect of the future research direction.

2. Preliminaries

In this section, an overview of key concepts and theories is provided, aimed at establishing a theoretical foundation for the following chapters. These concepts include fuzzy measures and Choquet integrals, D-S evidence theory, and the application of quantum probability theory in decision making.

2.1. Fuzzy measures and 2-additive Choquet integrals

Definition 1 [37]. Let $C = \left\{ c_1, c_2, \cdots, c_n \right\}$ be a nonempty finite set and $F(C)$ be a power set of C if the set function $\mu : F(C) \rightarrow [0,1]$ satisfies the following conditions:

(1) $\mu(\emptyset) = 0,\ \mu(C) = 1.$
(2) For arbitrary $S, Q \subseteq C,\ S \subseteq Q \Rightarrow \mu(S) \leq \mu(Q)$.

Then μ is a fuzzy measure defined on $(C, F(C))$, $\mu(c_j)$ represents the importance of $\{c_j\}$ itself, $\mu(S)$ and $\mu(Q)$ represent the importance of attribute sets S and Q.

Definition 2 [37]. For any $Q \in F(C)$, the Mobius transform of $\mu(Q)$ can be defined as:

$$m_\mu(Q) = \sum_{S \subseteq Q} (-1)^{|Q|-|S|} \mu(S). \tag{1}$$

Where $|Q|$ and $|S|$ represent the number of elements in the attribute sets Q and S respectively, that is, the cardinality of the sets. While $\mu(Q)$ and $m_\mu(Q)$ can be converted to each other. For any $Q \in F(C)$, $\mu(Q)$ can be represented by the Mobius expression:

$$\mu(Q) = \sum_{S \subseteq Q} m_\mu(S). \tag{2}$$

The overall importance of attribute c_j can be measured by Shapley value $\emptyset_{m_\mu}(c_j)$, which satisfies $\sum_{j=1}^{n} \emptyset m_\mu(c_j) = 1$. The Shapley value of attribute c_j is derived using the Mobius transformation as follows.

$$\emptyset_{m_\mu}(c_j) = \sum_{S \subseteq C \setminus \{c_j\}} \frac{1}{|S|+1} m_\mu(S \cup \{c_j\}), j = 1, \dots, n. \tag{3}$$

When $A = \left\{c_j\right\}$, I_j is called as the Shapley value of the attribute c_j, it is the average importance of the attribute c_j relative to other attributes.

Function f is a non-negative real-valued function defined on the set C, the set C represents the set of attributes c_j, the set $\bar{C}$ represents the complement of the set C. The single-attribute indicator I_j and the attribute-interaction indicator I_{jp} must adhere to the principles of monotonicity and regularity, which are defined as follows.

$$-2I_j \leq \sum_{c_p \in \bar{C} \setminus \{c_j\}} I_{jp} - \sum_{c_p \in C} I_{jp} \leq 2I_j, j = 1, 2, \cdots, n, p = 1, 2, \cdots, n. \tag{4}$$

When $A = \left\{c_i, c_j\right\}$, I_{ij} is called as the interaction index between attribute c_i and attribute c_j. If $I_{ij} > 0$, c_i and c_j are complementary promotion; if $I_{ij} < 0$, c_i and c_j are counteracting inhibition; if $I_{ij} = 0$, c_i and c_j are independent of each other. Based on the Shapley values of each attribute and the interaction indicators between attributes, the 2-additive Choquet integral formula is defined as follows [38]:

$$C_\mu(f) = \sum_{j=1}^{n} I_j f(c_j) - \frac{1}{2} \sum_{\{c_j, c_p\} \subset C} I_{jp} \left| f(c_j) - f(c_p) \right|. \tag{5}$$

2.2. D-S evidence theory

D-S evidence theory is a theoretical framework for integrating evidence from multiple sources and facilitating decision-making. The fundamental concept of this framework derives from the uncertainty in the evolution and dynamics of phenomena. Unlike classical Bayesian probability, it introduces a constructive probability function, termed a confidence function, which is capable of better representing incomplete information and subjective uncertainty information.

Definition 3 [39]. For any decision problem, all possible results are represented by the set Θ, $\Theta = \{\theta_1, ..., \theta_n\}$ is a non-empty finite set, and the elements are mutually exclusive, then Θ is called as the frame of discernment. 2^Θ is the power set of the frame of discernment Θ, representing all possible sets in Θ. The number of elements is 2^n, which can be expressed as $2^\Theta = \{\emptyset, \{\theta_1\}, ..., \{\theta_n\}, \{\theta_1, \theta_2\}, ..., \Theta\}$, event T is contained in Θ, each element in the power set 2^Θ represents an event in event T, that is, 2^Θ is all possible subsets of event T, $m(T)$ represents the basic probability assignment (BPA) value corresponding to event T, the extent to which the evidence supports the occurrence of event T.

Definition 4 [39]. Let Θ be the frame of discernment, and the basic probability assignment is a mapping $m : 2^\Theta \rightarrow [0, 1]$, which satisfies:

$$\begin{cases} m(\emptyset) = 0, \\ \sum\limits_{T \in \Theta} m(T) = 1, \\ 0 \leq m(T) \leq 1. \end{cases} \tag{6}$$

Definition 5 [39]. Let $m_1(B)$ and $m_2(C)$ be two basic probability assignment on the same frame of discernment Θ, and the fusion rules of D-S evidence theory are defined as follows.

$$m(T) = \begin{cases} 0, T = \emptyset, \\ \dfrac{\sum_{B \cap C = T, \forall B, C \subseteq \Theta} m_1(B) m_2(C)}{1-K}, T \neq \emptyset. \end{cases} \tag{7}$$

$$K = \sum_{B \cap C = \emptyset, \forall B, C \subseteq \Theta} m_1(B) m_2(C). \tag{8}$$

Where K is the conflict coefficient, indicating the degree of conflict between m_1 and m_2, and $0 \leq K \leq 1$. When $K = 0$, there is no conflict between m_1 and m_2. When $K = 1$, there is a complete conflict between m_1 and m_2.

2.3. Quantum probability theory

In contrast to classical probability theory, in quantum probability theory, all possible events can be viewed as subspaces of a Hilbert space. The Hilbert space is an extension of Euclidean space, formed by a complex vector space spanned by a set of orthogonal basis vectors $V = \{|V_1\rangle, \ldots, |V_n\rangle\}$. In the Hilbert space, all events are defined in terms of quantum superposition states $|D\rangle$, which can be represented as linear combinations of orthogonal basis vectors [40].

$$\begin{aligned} |D\rangle &= \langle V_1 \mid D\rangle |V_1\rangle + \langle V_2 \mid D\rangle |V_2\rangle + \cdots + \langle V_n \mid D\rangle |V_n\rangle \\ &= \psi_1 |V_1\rangle + \psi_2 |V_2\rangle + \cdots + \psi_n |V_n\rangle \\ &= \varphi_1 e^{i\theta_1} |V_1\rangle + \varphi_2 e^{i\theta_2} |V_2\rangle + \cdots + \varphi_n e^{i\theta_n} |V_n\rangle. \end{aligned} \tag{9}$$

Furthermore, using the complex number $\psi_k, \varphi_k e^{i\theta_k} (k = 1, 2, \ldots, n)$ to describe the probability amplitude of state $V_k (k = 1, 2, \ldots, n)$, where $(\varphi_k)^2 \in [0, 1]$, $\sum_{k=1}^{n} (\varphi_k)^2 = 1$. The term $e^{i\theta_k}$ is used to describe the phase of this complex wave function. According to the Born's rule, $(\varphi_k)^2$ is directly related to the observed probability of that state:

$$\begin{aligned} P(V_k) &= \left|\psi_k\right|^2 = |\varphi_k e^{i\theta_k}|^2 \\ &= (\varphi_k e^{i\theta_k}) \times (\varphi_k e^{-i\theta_k}) \\ &= \varphi_k e^{i\theta_k} \times \varphi_k e^{-i\theta_k} = (\varphi_k)^2 \quad (k = 1, 2, \ldots, n). \end{aligned} \tag{10}$$

2.4. Decision-making in quantum probability theory

In the traditional decision-making framework, it is usually assumed that the individual will evaluate all possible options and make a theoretically optimal decision based on these evaluations. However, quantum probability theory offers a whole new perspective, especially when it comes to non-classical probability calculations and how to make decisions in the face of uncertainty and complexity. Fig. 1 uses a Markov link path diagram to simulate changes in the state of the decision path. This graph intuitively shows the role of classical probability and quantum probability in the decision-making process.

In Fig. 1(a), according to the total probability principle in classical probability theory, P(A) can be calculated according to the total probability theorem:

$$P(A) = P(G)P(A|G) + P(B)P(A|B). \tag{11}$$

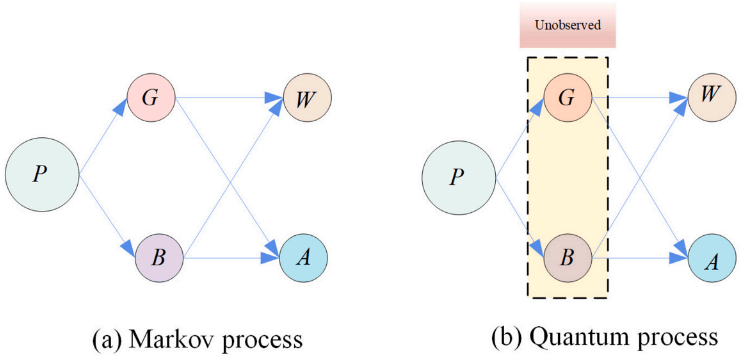

Fig. 1. Markov chain path diagram.

In quantum theory, the amplitude of the transition probability between states can be defined by the inner product of these two states. For example, in Fig. 1(b), the transition of states $P \to G$ can be represented by the Dirac symbol$\langle G|P\rangle$, where the amplitudes are complex rather than real numbers and the corresponding classical probability values are equal to the square of the magnitude of the amplitudes, i.e. $|\langle G|P\rangle|^2$ and $|\langle A|G\rangle|^2$. In this decision path, the results of quantum theory are consistent with those of classical theory. However, the probability calculation from $P \to A$ is more complicated and can be expressed similar to the total probability formula. For comparison, a Euler transformation in $\langle G|P\rangle \cdot \langle A|G\rangle$ and $\langle B|P\rangle \cdot \langle A|B\rangle$ is made to reveal the difference in the calculation of probabilities in these two cases:

$$
\begin{aligned}
P(A) = |\langle A|P\rangle|^2 &= |\langle G|P\rangle \cdot \langle A|G\rangle + \langle B|P\rangle \cdot \langle A|B\rangle|^2 \\
&= (\sqrt{P(G)P(A|G)} \cdot e^{i\alpha} + \sqrt{P(B)P(A|B)} \cdot e^{i\beta}) \\
&\quad \cdot (\sqrt{P(G)P(A|G)} \cdot e^{-i\alpha} + \sqrt{P(B)P(A|B)} \cdot e^{-i\beta}) \\
&= P(G)P(A|G) + P(B)P(A|B) + \\
&\quad 2\sqrt{P(G)P(A|G)P(B)P(A|B)} \cdot \cos(\alpha - \beta).
\end{aligned} \tag{12}
$$

Compared with classical probability theory, the formula of quantum probability has one more key component: the interference term $2\sqrt{P(G)P(A|G)P(B)P(A|B)} \cdot \cos(\alpha - \beta)$. This interference term is produced by the interaction of two different probability amplitudes. In quantum theory, this interference reflects the superposition and interaction of quantum states, which does not exist in classical probability theory. The existence of interference term is one of the key factors that distinguish quantum decision process from classical probabilistic decision process.

3. Multi-attribute quantum group decision model based on D-S evidence theory and Choquet integral

In this section, we construct an innovative quantum Bayesian framework to simulate interference phenomena in multi-attribute group decision-making. This framework can more precisely capture and address the intricate interactions within the decision-making environment, offering a novel perspective and a methodological foundation for studying interactions among attributes and interference of opinions in the decision-making process.

Suppose the multi-attribute group decision problem includes decision group D, alternative set A' and attribute set C. Where, decision group $D = \{d_1, d_2, \cdots, d_s\}$, alternative group $A' = \{a_1, a_2, \cdots, a_m\}$, attribute set $C = \{c_1, c_2, \cdots, c_n\}$. r_{ij}^k are the evaluation value given by decision maker d_k about alternative a_i upon attribute c_j, the weight of attribute c_j is $\omega_j, j = 1, ..., n$, and the weight of decision maker d_k is $\lambda_k, k = 1, ..., s$.

3.1. Determination of decision information

1) Evaluation information collection. The decision matrix is represented as $R^{(k)} = (r_{ij}^k)_{m\times n}$, where

$$
R^{(k)} = \begin{pmatrix} r_{11}^k & r_{12}^k & \cdots & r_{1n}^k \\ r_{21}^k & r_{22}^k & \cdots & r_{2n}^k \\ \vdots & \vdots & \ddots & \vdots \\ r_{m1}^k & r_{m2}^k & \cdots & r_{mn}^k \end{pmatrix}, k = 1, ..., s. \tag{13}
$$

2) Evaluation information matrix normalization. Since the evaluation values under different attributes have different dimensions or orders of magnitude, the incommensurability of data will be caused. In order to eliminate this impact, the data need to be normalized, and the normalized form $\tilde{r}_{ij}^k$ is computed as follows:

$$\tilde{r}_{ij}^k = \begin{cases} \dfrac{r_{ij}^k - \min_i r_{ij}^k}{\max_i r_{ij}^k - \min_i r_{ij}^k}, c_j \text{for benefit.} \\ \dfrac{\max_i r_{ij}^k - r_{ij}^k}{\max_i r_{ij}^k - \min_i r_{ij}^k}, c_j \text{for cost.} \end{cases} \tag{14}$$

3.2. Determination of attribute weights based on Choquet integral and D-S evidence theory

3.2.1. The attributes' Choquet score considering the interactive effects

Attributes often show interdependence and correlation. In order to describe the interaction between attributes, 2-Chocuet integral is used to score the importance of attributes.

1) During the process of recognizing fuzzy measures, a notable advantage lies in the fact that decision-makers are not required to provide a subjective assessment of the overall utility associated with each alternative. When it comes to recognizing 2-additive fuzzy measures, the method utilizes an optimization function to perform this process [37]. This function is formulated using the principle of minimizing variance to precisely capture and quantify interactions among multiple attributes in the decision-making process. This method not only improves the efficiency and accuracy of fuzzy measure recognition but also enhances its practicality and reliability in complex decision analysis. Where, $|C|$, $|S|$ and $|M|$ represent the numbers of elements contained in attribute sets C, S and M. The first constraint ensures that the measure remains non-negative when evaluating the contribution of any individual element or an empty set. The second constraint is the sum of these subsets must equal 1 for subsets comprising at least two elements.

$$\begin{aligned} &\min \frac{1}{n} \sum_{c_j \in C} \sum_{M \subseteq C \setminus \{c_j\}} \frac{(|C|-|M|-1)!|M|!}{|C|!} \left[\sum_{S \subseteq M} m_\mu (S \cup \{c_j\}) - \frac{1}{n} \right]^2 \\ &s.t. \begin{cases} \sum\limits_{S \subseteq M, |S| \le 1} (S \cup \{c_j\}) \ge 0, \forall c_j \in C, \forall M \subseteq C \setminus \{c_j\}, \\ \sum\limits_{S \subseteq M, 0<|S| \le 2} m_\mu (S) = 1. \end{cases} \end{aligned} \tag{15}$$

2) The overall significance of each attribute is employed by Shapley value.

$$\emptyset_{m_\mu}(c_j) = \sum_{S \subseteq C \setminus \{c_j\}} \frac{1}{|S|+1} m_\mu (S \cup \{c_j\}), j = 1, ..., n. \tag{16}$$

3) Based on the monotonicity and regularity of Choquet integral, the objective optimization model is used to solve the attribute interaction index. The following objective optimization model is constructed to solve the interaction index I_{pt}.

$$\begin{aligned} &\max \sum_{j=1}^{n} \left| C_u(c_j) \right| + \xi \sqrt{\frac{1}{n-1} \sum_{j=1}^{n} \left(C_u(c_j) - \bar{C}_u(c_j) \right)^2} \\ &s.t. \begin{cases} -2I_p \le \sum\limits_{c_p \in \bar{C} \setminus \{c_t\}} I_{pt} - \sum\limits_{c_p \in C} I_{pt} \le 2I_p, j = 1, 2, \cdots, n, \\ \bar{C}_u(c_j) = \frac{1}{n} \sum\limits_{j=1}^{n} C_u(c_j), \\ -1 \le I_{pt} \le 1, \{c_p, c_t\} \subset C, \\ \xi > 0. \end{cases} \end{aligned} \tag{17}$$

Where $\xi > 0$ is the adjustment coefficient of the term where the standard deviation is located. In order to avoid loss of generality, $\xi = 0.5$ is taken to ensure that the order of magnitude of each component of the objective function is the same and to avoid the problem of information loss caused by too large difference in order of magnitude.

Theorem 1. *The Model (17) has an optimal solution.*

Proof. Firstly, since the objective function contains absolute value and square root function, both of which have continuous properties, it can be inferred that the entire objective function is also continuous. Secondly, considering that the values of each element in the information evaluation matrix and the Choquet integral are in a bounded state, the objective function is also guaranteed to be bounded. On this basis, the constraints of the model constitute a closed set. Since the variables and are confined to closed intervals, the set defined by these constraints is naturally also closed.

According to Weierstrass approximation theorem [41], if a continuous function is defined on a closed and bounded set, then the function must reach its maximum and minimum values on that set. Considering that the objective function of this model is continuous and bounded, and the constraints are defined in a closed set, and the model is a single objective optimization problem, the feasible domain of this model exists and is bounded.

Based on the above analysis, considering the continuity and boundedness of the objective function and the closed set defined by the constraints, it can be seen that model (17) has an optimal solution. □

In multi-attribute group decision analysis, the concept of an optimal solution is paramount. This solution epitomizes an ideal decision-making strategy under specific conditions. It encompasses a comprehensive assessment that considers the significance of each attribute, as well as the dynamic interplay among these attributes. Such a solution is instrumental in harmonizing the diverse characteristics and their interrelations, aiming to realize the most favorable overall performance. This approach ensures a balanced and nuanced decision-making process, reflective of the complexity and multifaceted nature of the attributes involved.

5) Calculate a score for each attribute based on a 2-additive Choquet integral.

$$C_\mu{}^k(c_j) = \sum_{i=1}^{m} I_i \tilde{r}_{ij}^k - \frac{1}{2} \sum_{\{a_i, a_q\} \subset A'} I_{iq} \left| \tilde{r}_{ij}^k - \tilde{r}_{qj}^k \right|, j = 1, \ldots, n, k = 1, \ldots, s. \tag{18}$$

3.2.2. Attribute weight information fusion based on D-S evidence theory

The paper employs D-S evidence theory not merely as a method for evidence aggregation but uses it to integrate disparate information sources. This integration process is crucial for handling evidence conflict, a common issue when combining information from varied sources. This approach significantly enhances the robustness and reliability of the decision-making process by effectively managing conflicts among evidence, thus extending the utility of D-S evidence theory beyond its conventional applications. And the paper elaborates on a systematic approach to first aggregate evidence sources using D-S evidence theory, then calculate attribute weights, and subsequently convert these weights into conditional probabilities for Bayesian networks. This sequential process exemplifies a comprehensive and methodical application of D-S evidence theory, culminating in a practical framework for decision-making that leverages the strengths of both D-S theory and Bayesian networks.

The frame of discernment of D-S evidence theory is constructed, which is a complete set Θ assigned by mutually exclusive events, expressed as: $\Theta = \{c_1, \ldots c_n\}$. Let the BPA function in the identification framework Θ be $m_k, k = 1, \ldots, s$ and its focal element be $c_j, j = 1, \ldots, n$. The mass function in the identification framework is defined as:

$$m_k(c_j) = \frac{C_\mu{}^k(c_j)}{\sum_{j=1}^{n} C_\mu{}^k(c_j)}, k = 1, \ldots, s, j = 1, \ldots, n. \tag{19}$$

Conflict factors are determined by fusing the panel's evidence.

$$K = \sum_{\{c_j\} \cap \{c_{j'}\} = \emptyset} m_k(c_j) m_{k'}(c_{j'}), k = 1, \ldots, s, j = 1, \ldots, n, k \neq k', j \neq j'. \tag{20}$$

Basic probability assignment based on evidence fusion.

$$m(c_j) = \frac{\sum_{\{c_j\} \cap \{c_{j'}\}} m_k(c_j) m_{k'}(c_{j'})}{1 - K}, j = 1, \ldots, n. \tag{21}$$

In actual decision environment, the decision-maker usually assigns the subjective weight vector $\alpha = (\alpha_1, \ldots, \alpha_n)$ to the attributes' set, then the comprehensive weight of the attributes is:

$$\omega_j = g\alpha_j + (1 - g) m(c_j), j = 1, \ldots, n. \tag{22}$$

Where, $0 \leq g \leq 1$, it is used to coordinate the importance of subjective or objective weights.

3.3. Decision makers' weights based on grey relative relational degree

Taking into account the probability of the first layer of the Bayesian network, the decision-makers' weights model are constructed based on the grey relative correlation degree between each expert's comprehensive value sequence and the comprehensive value sequence integrating all expert opinions.

Each expert's comprehensive value sequence is deemed as reference sequence.

$$\hat{R}^k = \left(\hat{r}_1^k, \hat{r}_2^k \ldots, \hat{r}_m^k\right), k = 1, \ldots, s. \tag{23}$$

Where,

$$\hat{r}_i^k = \frac{1}{n}\sum_{j=1}^{n} \tilde{r}_{ij}^k, i = 1, ..., m, k = 1, ..., s. \tag{24}$$

The maximum value of the comprehensive evaluation of each alternative by all decision makers is calculated.

$$\hat{r}_i^+ = \max_{1\le k\le s} \hat{r}_i^k, i = 1, 2, \ldots, m. \tag{25}$$

The sequence being referenced is:

$$\hat{R}^+ = \left(\hat{r}_1^+, \hat{r}_2^+, \ldots, \hat{r}_m^+\right). \tag{26}$$

Then:

$$\left|s_k\right| = \left|\sum_{i=2}^{m-1} \hat{r}_i^k + \frac{1}{2}\hat{r}_m^k\right|. \tag{27}$$

$$\left|s_o\right| = \left|\sum_{i=2}^{m-1} \hat{r}_i^+ + \frac{1}{2}\hat{r}_m^+\right|. \tag{28}$$

$$\left|s_k - s_o\right| = \left|\sum_{i=2}^{m-1} \left(\hat{r}_i^k - \hat{r}_i^+\right) + \frac{1}{2}\left(\hat{r}_m^k - \hat{r}_m^+\right)\right|. \tag{29}$$

The grey relative relational degree is calculated as:

$$\varepsilon_{ok} = \frac{1}{1 + \left|s_k - s_o\right|}. \tag{30}$$

Theorem 2. *The closer the behavior sequence $\hat{R}^k$ is to the reference sequence $\hat{R}^+$, the greater the correlation ε_{ok} between them.*

Proof. In grey relational analysis, the relational grade ε_{ok} represents the relationship between the behavior sequence $\hat{R}^k$ and the reference sequence $\hat{R}^+$. The measurements s_k and s_o respectively represent the behavior sequence $\hat{R}^k$ and the reference sequence $\hat{R}^+$. The calculation of this relational grade is based on the degree of difference between the sequences, where the degree of difference is expressed as $\left|s_k - s_o\right|$.

Firstly, the degree of difference $\left|s_k - s_o\right|$ is a key indicator for measuring the difference between the behavior sequence $\hat{R}^k$ and the reference sequence $\hat{R}^+$. When the elements $\hat{r}_i^k$ in the sequence $\hat{R}^k$ are closer to the corresponding elements $\hat{r}_i^+$ in the reference sequence $\hat{R}^+$, the value of this difference $\left|s_k - s_o\right|$ becomes smaller. That is, the higher the degree of proximity between the sequence $\hat{R}^k$ and $\hat{R}^+$, the lower their degree of difference.

On this basis, it can be observed that the value of the relational grade is the reciprocal of $1 + \left|s_k - s_o\right|$. Therefore, when the value of $\left|s_k - s_o\right|$ decreases, the difference between the sequence $\hat{R}^k$ and $\hat{R}^+$ decreases, the denominator $1 + \left|s_k - s_o\right|$ also decreases, thereby the value of ε_{ok} increases. This indicates that the closer the sequence $\hat{R}^k$ is to the reference sequence $\hat{R}^+$, the greater the relational grade ε_{ok} between them. □

Finally, the weight of decision maker is calculated as:

$$\lambda_k = \frac{\varepsilon_{ok}}{\sum_{k=1}^{s} \varepsilon_{ok}}, k = 1, ..., s. \tag{31}$$

3.4. Quantum probabilities of alternatives based on interference effects

Meteorological disaster emergency rescue operations are complex, highly dynamic, and involve multiple attributes and decision-makers with varying preferences and judgments. These operations require rapid, accurate, and objective decision-making to minimize the impact of disasters. Recognizing these challenges, our model is tailored to address the unique needs of meteorological disaster emergency response.

By addressing the specific characteristics of meteorological disaster emergency rescue, our MAQGDM provides a comprehensive and nuanced approach to decision-making in this critical context. The model's integration of quantum decision-making principles with traditional decision-making methods offers a novel and effective tool for emergency response planning and execution. And the detailed model design thoughts are as follows.

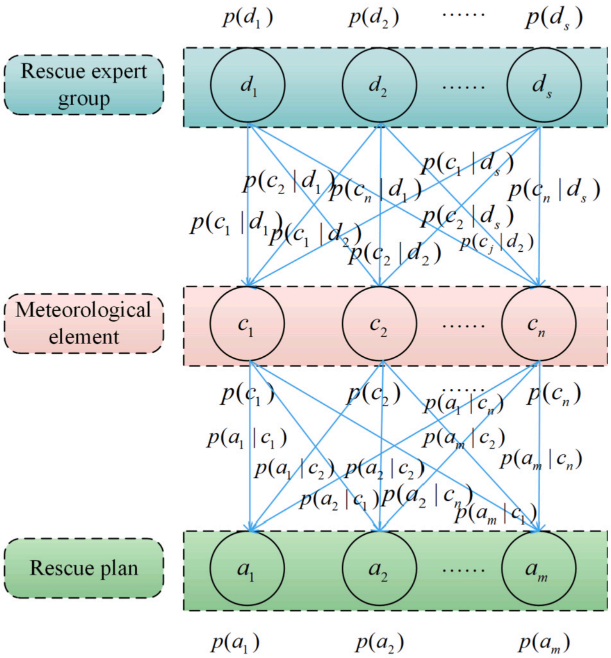

Fig. 2. Bayesian networks in the MAQGDM problem of meteorological disaster.

1) Interaction Handling: Our model utilizes the 2-addition Choquet integral based on Mobius transformation to effectively handle the interactions between decision attributes. This approach significantly reduces subjective biases, enhancing the objectivity and accuracy of decision-making.
2) Decision Maker Weight Determination: To account for the varying influence of decision-makers, we employ the grey relational degree to determine their weights. These weights are then integrated as base node probabilities within a Bayesian network framework, ensuring a robust representation of decision-makers' perspectives.
3) Attribute Value Integration: Considering the individual differences among decision-makers, our model adopts D-S evidence theory to integrate interactions between different attribute values. This integration synthesizes the final attribute weights and combines them with the information evaluation matrix, leading to the derivation of conditional probabilities within the quantum framework.
4) Quantum Probability Computation: The model computes the quantum probabilities of alternatives based on Bayesian principles. The ranking of alternatives is determined by these probabilities, enabling the selection of the most suitable emergency plan.

Within the quantum decision-making framework, this study employs a three-tier Bayesian network to simulate the MAQGDM process. The first tier of the network represents different decision-making groups, where each decision-maker may be influenced by the information or preferences of others. The second tier is the attribute layer, which depicts the various attributes associated with multi-attribute decision-making problems. The third tier encompasses all possible rescue options, serving as the alternative solutions layer. This framework intricately details the interference effects among different attributes and between decision-makers. Specifically, given the characteristics of meteorological disasters, the first layer of the Bayesian network can be interpreted as the rescue expert group, the second layer as meteorological elements, and the third layer as potential rescue plans. Fig. 2 illustrates this process.

Let decision makers' weights be the first layer probability of Bayesian networks.

$$p(d_k) = \lambda_k, k = 1, \dots, s. \tag{32}$$

The ratio of the synthetic attribute value of each decision maker to the total synthetic attribute value is used as the conditional probability, which is used as a tool to connect the decision group with the alternatives in BN.

$$p(a_i|d_k) = \frac{\sum_{j=1}^{n} \omega_j \tilde{r}_{ij}^k}{\sum_{i=1}^{m} \sum_{j=1}^{n} \omega_j \tilde{r}_{ij}^k}, i = 1, \dots, m, k = 1, \dots, s. \tag{33}$$

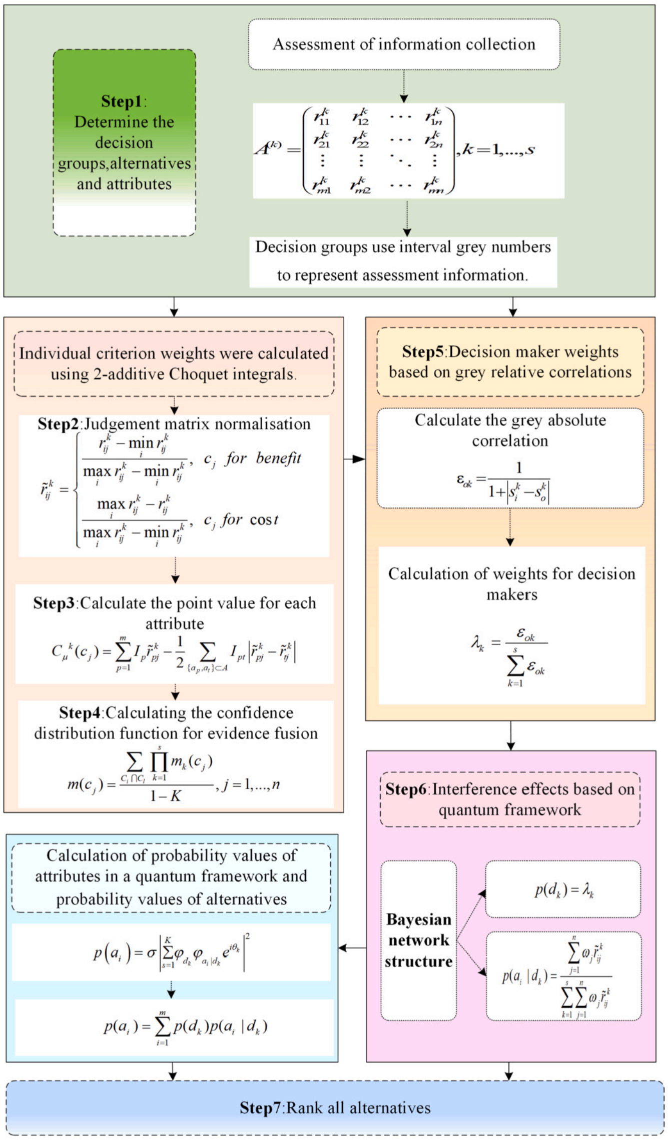

Fig. 3. Quantum group decision making steps.

The probability of each alternative is calculated as:

$$p(a_i) = \sum_{k=1}^{s} p(d_k)p(a_i|d_k) + 2\sum_{k=1}^{s}\sum_{k'=k+1}^{s} \sqrt{p(d_k)p(a_i|d_k)p(d_{k'})p(a_i|d_{k'})}\cos(\theta_k - \theta_{k'}), i = 1, \ldots, m. \tag{34}$$

The Angle between the opinions of different decision makers is analyzed, and the probability of alternative is obtained, then the alternatives are ranked by the values.

4. MAQGDM model steps and comparison

4.1. Decision steps

The modeling steps are as follows, which are illustrated in Fig. 3.

Table 1
The comparison of the proposed method and previous decision models.

Models	Information	Attribute weights	DMs weights	Interference value measurement	Measure interference Angle	Ranking
3WMADM [42]	Fermatean fuzzy set	Unknown	—	None	No	Yes
DM [43]	Type-2 fuzzy set	—	—	None	No	No
MADM [44]	T-spherical fuzzy sets	Known	Known	None	No	Yes
MCDM [45]	PIVq-ROHFS	Unknown	—	None	No	Yes
MCGDM [17]	Probabilistic linguistic term sets	Unknown	Unknown	Similarity vector heuristic algorithm	Yes	Yes
MCGDM [19]	Linguistic term set	Known	Known	Phase Angle change calculation	Yes	Yes
MADM [18]	Crisp number	Known	—	Standardized information entropy transformation	No	Yes
Proposed model	Crisp number	Unknown	Unknown	Phase Angle change calculation	Yes	Yes

Step 1: Determine the evaluation information. Each decision maker $d_k(k=1,...,s)$ evaluates attribute $c_j(j=1,2,...,n)$ about alternative $a_i(i=1,...,m)$, and obtains the evaluation information matrix $R^{(k)}=(r_{ij}^k)_{m\times n}(k=1,2,...,s)$.

Step 2: Normalize the evaluation information matrix. Each evaluation information matrix $R^{(k)}=(r_{ij}^k)_{m\times n}(k=1,2,...,s)$ is standardized according to Eq. (14), and the standardized evaluation information matrix $\tilde{R}^{(k)}=(\tilde{r}_{ij}^k)_{m\times n}(k=1,2...,s)$ is obtained.

Step 3: Apply the 2-Choquet integral to calculate the importance of the attribute set. Firstly, a fuzzy measure optimization model based on least variance method is constructed according to Eq. (15). Then, the Shapley value I_p of each attribute is determined by Eq. (16). In addition, in order to solve the interaction index $I_{pt}, p=1,2,\cdots,m, t=1,2,\cdots,m$ between the two attributes, a target optimization model Eq. (17) is constructed, and finally, the integral value $C_\mu(c_j)(j=1,2,...,n)$ of each attribute is calculated by using Eq. (18).

Step 4: Determine attribute weights based on D-S theory. The mass function value based on the Choquet integral values of each attribute is obtained by Eq. (19). The conflict factor when combining the opinions of different expert groups is obtained by Eq. (20). The basic probability assignment (BPA) is obtained by Eq. (21). Finally, the comprehensive weight of each attribute is calculated according to Eq. (22).

Step 5: Decision maker weight determination based on grey relative correlation degree. The weight of each decision maker is calculated from the Eq. (23)-Eq. (31).

Step 6: Determine the quantum Bayesian framework. The weight $\lambda_k, k=1,2,...,s$ of the decision group is determined as the node probability $p(d_k), k=1,2,...,s$ in the first layer of the Bayesian network. The conditional probability $p(a_i|d_k), i=1,2,...,m, k=1,2,...,s$ in Bayesian network is obtained from Eq. (33).

Step 7: Alternative evaluation considering quantum interference effects. The quantum probability $p(a_i), i=1,2,...,m$ of each alternative is obtained by Eq. (34), and the alternatives are sorted according to the quantum probabilities.

4.2. Comparison of models

In this section, the paper analyzes the differences between the proposed new model and previous decision-making models from various perspectives, thereby demonstrating the superiority of the model introduced in this paper. The differences between the proposed method and other reference models are illustrated in the Table 1.

It is seen from the Table 1 that most current models do not consider the interference effects between decision groups, and the determination of attribute weights and decision group weights is not always achieved through objective calculations. By contrast, this paper introduces a new method to calculate attribute weights and reflects the interference among decision groups by incorporating quantum interference. The main features of the model presented in this paper are as follows.

The quantum decision theory is extended to MAGDM problems. It accounts for the influence of belief interference among decision-makers at different decision stages, with the final outcomes obtained through the aggregation of attribute probabilities and the aggregation of substitute probabilities in the Quantum-Like Bayesian Network.

Attribute weights are determined through evidence theory and the 2-Choquet integral, and decision group weights are ascertained using grey relational analysis, ensuring objectivity.

From the perspective of quantum uncertainty, the interference effects of decision groups are discussed by considering changes in the phase angle, observing the resultant variations in the ranking of the final options.

5. Case analysis

In this section, we employ a meteorological disaster emergency case to validate the feasibility of the proposed MAQGDM model, demonstrating it effectively addresses the interaction between decision makers and the correlation between attributes.

5.1. Background introduction

In July 2021, Zhengzhou and its surrounding areas in Henan Province, China were hit by unusually strong rainfall, resulting in widespread secondary disasters such as urban waterlogging, flash floods, mudslides and house collapses, which seriously threatened local social stability and people's lives. In this context, it becomes particularly important to develop a comprehensive emergency response plan, which requires accurate assessment of the scale and impact of the disaster, effective deployment of relief resources, and efficient communication mechanisms to ensure the safe evacuation of personnel and the provision of necessary medical and livelihood support. There are five different emergency rescue plans $a_1, a_2, \cdots, a_5$ containing flood discharge, resident transfer, material distribution, rescue force deployment, etc, and considers the following five disaster attributes to measure the ranking of the plans.

1) Casualties c_1: This attribute measures the number of casualties, which includes the number of injuries and deaths. The unit is "number of people".
2) Economic loss c_2: This attribute reflects the direct impact of the event on the economy, and the unit of measurement of loss is "millions of yuan", which includes the loss of economic activities such as property damage and production stagnation.
3) Infrastructure resilience c_3: This index employs a scoring system ranging from 0 to 10 to evaluate the resilience of critical infrastructure. These advanced facilities include medical centers, emergency services, critical communications base stations, etc. The higher the score, the more resilient the facility is and the faster it can return to normal operations.
4) Traffic disruption area c_4: This index gauges the impact of disasters on the transportation network, the unit is "acres". It assesses the degree of disruption to roads, railways, or other modes of transport caused by disasters like floods and mudslides.
5) Communication interruption time c_5: This attribute is used to measure the duration of communication network outages caused by disasters, the unit is "hours".

There are three expert group d_1, d_2, d_3 to evaluate the emergency plans. In the process of emergency rescue, there is usually a correlation between disaster attributes. Meanwhile, decision makers' evaluation and selection of plans will also be interfered with and influenced by other decision makers, so, we use the quantum group decision model to select the emergency plans.

5.2. Decision steps

Step 1: Determine the evaluation information. The following matrices represent the evaluation values of the three groups of experts. The data was based on the "Survey Report of the July 20 Torrential Rain Disaster in Zhengzhou, Henan Province," representing actual disaster-related data. These datasets are publicly accessible through the Ministry of Emergency Management's official website at https://www.mem.gov.cn/.

$$R^{(1)} = \begin{pmatrix} 360 & 320 & 8 & 3000 & 30 \\ 400 & 340 & 6 & 3400 & 40 \\ 390 & 380 & 4 & 2800 & 35 \\ 470 & 370 & 3 & 3600 & 46 \\ 340 & 430 & 5 & 4000 & 38 \end{pmatrix},$$

$$R^{(2)} = \begin{pmatrix} 340 & 420 & 7 & 3200 & 32 \\ 390 & 390 & 6 & 3800 & 38 \\ 420 & 460 & 5 & 4200 & 47 \\ 540 & 330 & 3 & 3000 & 56 \\ 360 & 360 & 4 & 2800 & 42 \end{pmatrix},$$

$$R^{(3)} = \begin{pmatrix} 370 & 370 & 6 & 4100 & 26 \\ 450 & 390 & 8 & 3500 & 20 \\ 460 & 430 & 5 & 3800 & 29 \\ 600 & 360 & 2 & 5020 & 35 \\ 410 & 340 & 7 & 4600 & 39 \end{pmatrix}.$$

Step 2: Normalization of evaluation information. Each individual information evaluation matrix $R^{(k)} = (r_{ij}^k)_{m\times n}$ is normalized to ensure data consistency and comparability. After normalization, the information evaluation matrix is:

$$\tilde{R}^{(1)} = \begin{pmatrix} 0.8462 & 1 & 1 & 0.8333 & 1 \\ 0.5385 & 0.8182 & 0.6 & 0.5 & 0.375 \\ 0.6154 & 0.4545 & 0.2 & 1 & 0.6875 \\ 0 & 0.5455 & 0 & 0.3333 & 0 \\ 1 & 0 & 0.4 & 0 & 0.5 \end{pmatrix},$$

$$\tilde{R}^{(2)} = \begin{pmatrix} 1 & 0.3077 & 1 & 0.7143 & 1 \\ 0.75 & 0.5385 & 0.75 & 0.2857 & 0.75 \\ 0.6 & 0 & 0.5 & 0 & 0.375 \\ 0 & 1 & 0 & 0.8571 & 0 \\ 0.9 & 0.7692 & 0.25 & 1 & 0.5833 \end{pmatrix},$$

$$\tilde{R}^{(3)} = \begin{pmatrix} 1 & 0.6667 & 0.6667 & 0.6053 & 0.6842 \\ 0.6522 & 0.4444 & 1 & 1 & 1 \\ 0.6087 & 0 & 0.5 & 0.8026 & 0.5263 \\ 0 & 0.7778 & 0 & 0 & 0.2105 \\ 0.8261 & 1 & 0.8333 & 0.2763 & 0 \end{pmatrix}.$$

Step 3: Calculate the 2-additive Choquet integral value for each attribute under every decision-maker. The Shapley value for single attributes and the interaction values for attributes by using Eq. (15)-Eq. (17), finally, the results are calculated using Eq. (18).

$$C_u^1(C) = [0.0047, 0.1898, 0.0310, 0.1399, 0.0886],$$

$$C_u^2(C) = [0.0804, 0.0876, 0.1133, 0.1136, 0.0719],$$

$$C_u^3(C) = [0.0833, 0.1287, 0.0318, 0.0870, 0.1154].$$

Step 4: Calculate the mass function value. On the basis of calculating the Choquet integral value, the mass function value for each attribute in each decision group is calculated using Eq. (19).

$$m_1(C) = (0.0104, 0.4181, 0.0683, 0.3081, 0.1952),$$

$$m_2(C) = (0.1722, 0.1877, 0.2427, 0.2434, 0.1540),$$

$$m_3(C) = (0.1867, 0.2884, 0.0713, 0.1950, 0.2586).$$

Then, we obtain the comprehensive probability distribution function using the BPA obtained from Eq. (21).

$$m(C) = (0.0072, 0.4862, 0.0254, 0.3142, 0.1670).$$

Suppose the attribute set has an initial subjective weight $\alpha = (0.1, 0.3, 0.2, 0.3, 0.1)$, let $g = 0.5$, the comprehensive weight of the attribute is calculated according to Eq. (22) as follows:

$$\omega_1 = 0.0536, \qquad \omega_2 = 0.3931, \qquad \omega_3 = 0.1127,$$

$$\omega_4 = 0.3071, \qquad \omega_5 = 0.1335.$$

Step 5: Decision maker weight determination based on grey relative correlation degree. By applying Eq. (23)-Eq. (31), we can obtain:

$$\lambda_1 = 0.3051, \qquad \lambda_2 = 0.3270, \qquad \lambda_3 = 0.3679.$$

Step 6: Determine quantum probabilities of the Bayes framework. The weights of the decision group are converted into the probability of the first node in the Bayesian network, which can be obtained:

$$p(d_1) = 0.3051, \qquad p(d_2) = 0.3270, \qquad p(d_3) = 0.3679.$$

The conditional probability in the Bayesian network can be calculated by Eq. (33) as follows:

$$p(a_1|d_1) = 0.3513, \qquad p(a_2|d_1) = 0.2322, \qquad p(a_3|d_1) = 0.2364,$$

$$p(a_4|d_1) = 0.1183, \qquad p(a_5|d_1) = 0.0618, \qquad p(a_1|d_2) = 0.2351,$$

$$p(a_2|d_2) = 0.1925, \qquad p(a_3|d_2) = 0.0509, \qquad p(a_4|d_2) = 0.2410,$$

$$p(a_5|d_2) = 0.2805, \qquad p(a_1|d_3) = 0.2397, \qquad p(a_2|d_3) = 0.2738,$$

$$p(a_3|d_3) = 0.1456, \qquad p(a_4|d_3) = 0.1198, \qquad p(a_5|d_3) = 0.2211.$$

Step 7: Using Eq. (34) to get the quantum probability of each emergency plans:

$$\begin{aligned} p(a_1) &= \left| \sum_{k=1}^{3} \sqrt{p(d_k)} \sqrt{p(a_1|d_k)} e^{i\Theta_k} \right| \\ &= 0.2722 + 0.18150\theta + 0.1944\beta + 0.1647\gamma, \end{aligned}$$

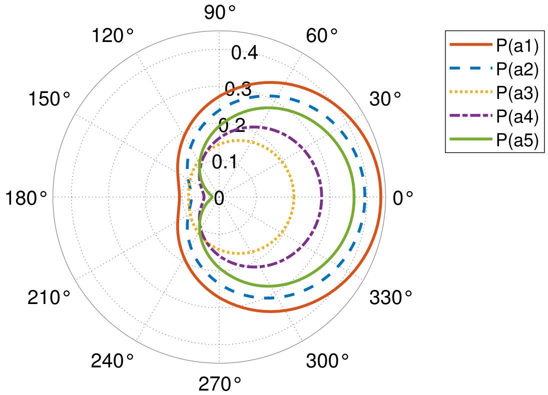

Fig. 4. Polar coordinate diagram of probability values of emergency plans changing with γ.

$$p(a_2)=\left|\sum_{k=1}^{3}\sqrt{p(d_k)}\sqrt{p(a_2|d_k)}e^{i\Theta_k}\right|$$
$$=0.2345+0.13336\theta+0.1689\beta+0.1593\gamma,$$
$$p(a_3)=\left|\sum_{k=1}^{3}\sqrt{p(d_k)}\sqrt{p(a_3|d_k)}e^{i\Theta_k}\right|$$
$$=0.1423+0.0693\theta+0.1243\beta+0.0597\gamma,$$
$$p(a_4)=\left|\sum_{k=1}^{3}\sqrt{p(d_k)}\sqrt{p(a_4|d_k)}e^{i\Theta_k}\right|$$
$$=0.1590+0.1067\theta+0.0798\beta+0.1179\gamma,$$
$$p(a_5)=\left|\sum_{k=1}^{3}\sqrt{p(d_k)}\sqrt{p(a_5|d_k)}e^{i\Theta_k}\right|$$
$$=0.1919+0.0832\theta+0.0783\beta+0.1727\gamma.$$

Each emergency plan is ranked according to its quantum probability score. It is assumed that there are no interference effects between decision groups d_1 and d_2, and between d_1 and d_3. Only interference effects between d_2 and d_3 are considered, so different phase angles are set to obtain probability values for the emergency plans.

In Fig. 4, we present the transformation of probability fractions in polar coordinates as varies γ from 0 to 2π. The polar coordinate system effectively illustrates the relationship between the function and changes in angle, particularly when the function's properties are angle-dependent. For datasets exhibiting periodicity or symmetry, polar plots offer an intuitive visual representation that accentuates these inherent properties. The following key observations can be made from the polar diagram:

1) The function image is closed at $\gamma=0$ and $\gamma=2\pi$, indicating that each probability function presents a clear periodicity.
2) When $\gamma=\frac{\pi}{2}$, there is no interference between d_2 and d_3, they are completely independent. Therefore, in this case, the result is the same as the normal weighted average fusion mode, and the final sort order is: $a_1 \succ a_2 \succ a_5 \succ a_4 \succ a_3$.
3) When $\gamma \in \left[0, \frac{\pi}{2}\right]$, there is positive interference between d_2 and d_3. Despite the drop in results, the positive effect results in a higher result than the independent case, and the final sort order is: $a_1 \succ a_2 \succ a_5 \succ a_4 \succ a_3$.
4) When $\gamma \in \left[\frac{\pi}{2}, \pi\right]$, there is negative interference between d_2 and d_3. On the one hand, the result drops, on the other hand, the negative effect leads to a lower result than in the independent case, when $\gamma=\pi$, all functions reach a local minimum, which means that there is completely negative interference between the decision groups in that range of angles, so the final sort order: $a_1 \succ a_3 \succ a_2 \succ a_4 \succ a_5$.

Then, the consistency and accuracy of the analysis can be further verified by comparing the conclusion of the polar diagram with Fig. 5 in the Cartesian coordinate system.

In Fig. 4 and Fig. 5, the ranking of the final choice remains largely consistent with the results obtained from the traditional decision model in most cases despite the presence of interference effects. In most cases, the most popular plans do not change due to interference effects even under some extremely negative conditions. At the same time, since most decision makers hold a positive or neutral attitude towards interference items, the conflict or uncertainty caused by interference effects is reduced. So, taking into

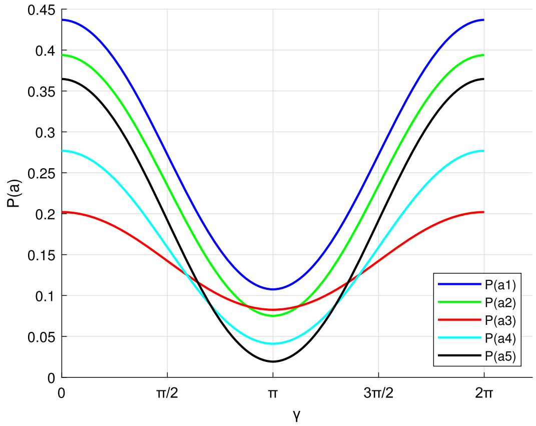

Fig. 5. Cartesian coordinate diagram of probability values of emergency plans changing with γ.

Table 2
The alternative probability values considering changes in γ and g.

γ	g	$p(a_1)$	$p(a_2)$	$p(a_3)$	$p(a_4)$	$p(a_5)$	Ranking order
	0	0.4185	0.3829	0.1744	0.3301	0.3639	$a_1 \succ a_2 \succ a_4 \succ a_5 \succ a_3$
	0.2	0.4259	0.3873	0.1858	0.3088	0.3642	$a_1 \succ a_2 \succ a_4 \succ a_5 \succ a_3$
0	0.4	0.4333	0.3916	0.1967	0.2875	0.3645	$a_1 \succ a_2 \succ a_4 \succ a_5 \succ a_3$
	0.6	0.4405	0.3960	0.2073	0.2663	0.3648	$a_1 \succ a_2 \succ a_4 \succ a_5 \succ a_3$
	0.8	0.4477	0.4002	0.2176	0.2452	0.3651	$a_1 \succ a_2 \succ a_4 \succ a_5 \succ a_3$
	1	0.4548	0.4045	0.2276	0.2241	0.3655	$a_1 \succ a_2 \succ a_4 \succ a_5 \succ a_3$
	0	0.2630	0.2294	0.1315	0.1881	0.1880	$a_1 \succ a_2 \succ a_4 \succ a_5 \succ a_3$
	0.2	0.2667	0.2315	0.1359	0.1764	0.1895	$a_1 \succ a_2 \succ a_5 \succ a_4 \succ a_3$
$\pi/2$	0.4	0.2704	0.2335	0.1402	0.1648	0.1911	$a_1 \succ a_2 \succ a_5 \succ a_4 \succ a_3$
	0.6	0.2741	0.2355	0.1445	0.1532	0.1927	$a_1 \succ a_2 \succ a_5 \succ a_4 \succ a_3$
	0.8	0.2778	0.2375	0.1487	0.1417	0.1944	$a_1 \succ a_2 \succ a_5 \succ a_3 \succ a_4$
	1	0.2814	0.2395	0.1529	0.1302	0.1960	$a_1 \succ a_2 \succ a_5 \succ a_3 \succ a_4$
	0	0.1075	0.0759	0.0887	0.0461	0.0121	$a_1 \succ a_3 \succ a_2 \succ a_4 \succ a_5$
	0.2	0.1075	0.0757	0.0860	0.0441	0.0149	$a_1 \succ a_3 \succ a_2 \succ a_4 \succ a_5$
π	0.4	0.1075	0.0754	0.0837	0.0421	0.0177	$a_1 \succ a_3 \succ a_2 \succ a_4 \succ a_5$
	0.6	0.1076	0.0751	0.0816	0.0401	0.0206	$a_1 \succ a_3 \succ a_2 \succ a_4 \succ a_5$
	0.8	0.1078	0.0748	0.0798	0.0382	0.0236	$a_1 \succ a_3 \succ a_2 \succ a_4 \succ a_5$
	1	0.1080	0.0746	0.0781	0.0362	0.0266	$a_1 \succ a_3 \succ a_2 \succ a_4 \succ a_5$

account personal preferences and interference effects, the ranking is $a_1 \succ a_2 \succ a_5 \succ a_4 \succ a_3$. In this case, considering the attribute interaction and opinion interference effect, the best emergency plan is a_1.

5.3. Sensitivity analysis

In this section, we analyze the influence of model parameter on the final calculation results and rankings. The parameter g reflects the balance of bias among the decision team when assigning subjective and objective weights. The larger parameter g is, the more emphasis is placed on subjective factors in determining attribute weights. The parameter g was systematically varied from 0 to 1 in increments of 0.2, enabling researchers to observe the impact of shifting weight allocation from entirely objective to entirely subjective on the ultimate decision outcomes.

In addition, the setting of phase Angle γ is also part of the study, it is set to 0, $\frac{1}{2}\pi$, π. In this paper, phase Angle is used to simulate the change of attitude of decision makers in different situations. By adjusting parameter value g and phase Angle γ, we can observe the change of decision result under different combination conditions. Given that θ, β, and γ parameters are symmetric in terms of their roles in the model, examining the sensitivity of one parameter serves to reflect the potential sensitivity of the others as well.

This research method significantly contributes to a deeper understanding of the interplay between subjective and objective factors within the group decision-making process and their consequential effects on the final ranking and selection outcomes. By employing parameter settings and result observation, it shows that effectively adjusting and optimizing multi-criteria group decision-making model can better adapt to different decision-making situations. The obtained results are presented in Table 2 and Fig. 6. Table 2 presents the ranking of alternatives in different γ and g scenarios, Fig. 6 shows how the probability scores of these alternatives change with γ and g, where, the phase angles γ are 0, $\pi/2$, π, and g increase by 0.2 each time between $[0, 1]$.

From Table 2, we know there are different ranking orders based on different γ and g, and we can conclude the following results.

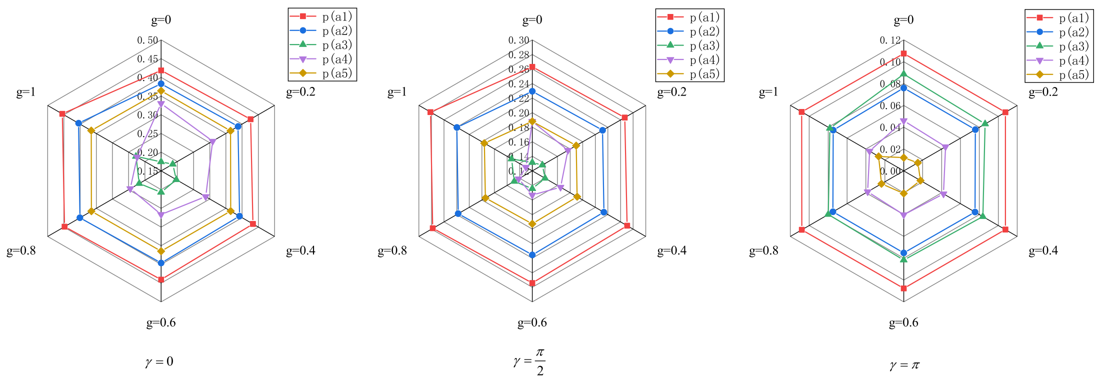

Fig. 6. Alternative probability changes based on changes in γ and g.

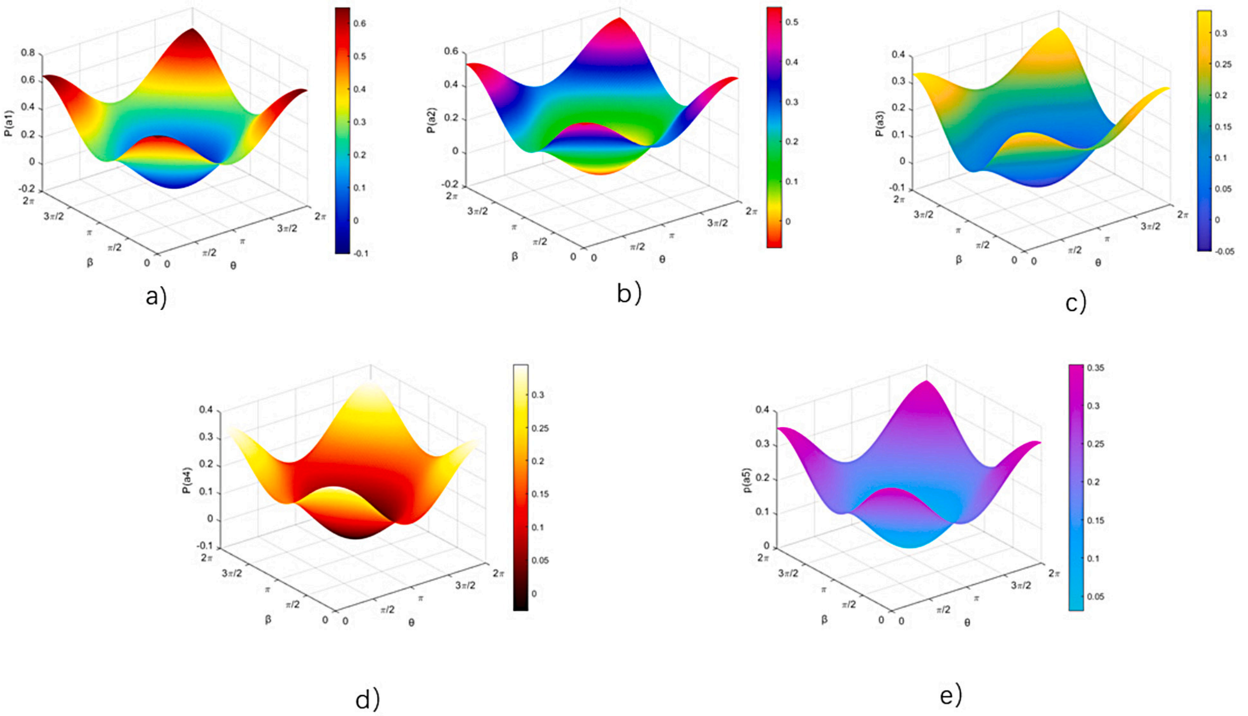

Fig. 7. Probability changes for alternatives based on changes in θ and β.

1) In the case of a fixed γ value, different g values will cause the probability scores of all alternatives to change, but the ranking usually remains stable.
2) The variation of probability scores of alternative solutions presents a fluctuating distribution, which is consistent with the characteristic of amplitude variation.
3) Phase Angle γ has a significant impact on the ranking of alternatives. For example, alternative a_1 is optimal when interference factors have a positive effect, and alternative a_3 is the worst choice. When interference factors have a negative impact, alternative a_1 becomes the best choice, but alternative a_5 is the worst choice.

It is found that the specific coefficient g has little effect on the ranking order, showing the insensitivity to this coefficient. In contrast, phase Angle γ is significantly sensitive to ranking order. These findings suggest that the proposed MAQGDM is both efficient and effective. This model is especially suitable for improving flexibility and stability in real decision making scenarios.

Then, we explore the mutual interference effects of different decision groups in the calculation and final ranking. Interference effects between d_1 and d_2, d_1 and d_3 are considered, while d_2 and d_3 are treated as independent. By using different phase angles θ and β, it is found that the probabilities of the alternatives present symmetric and dynamic characteristics. These results are shown in Fig. 7.

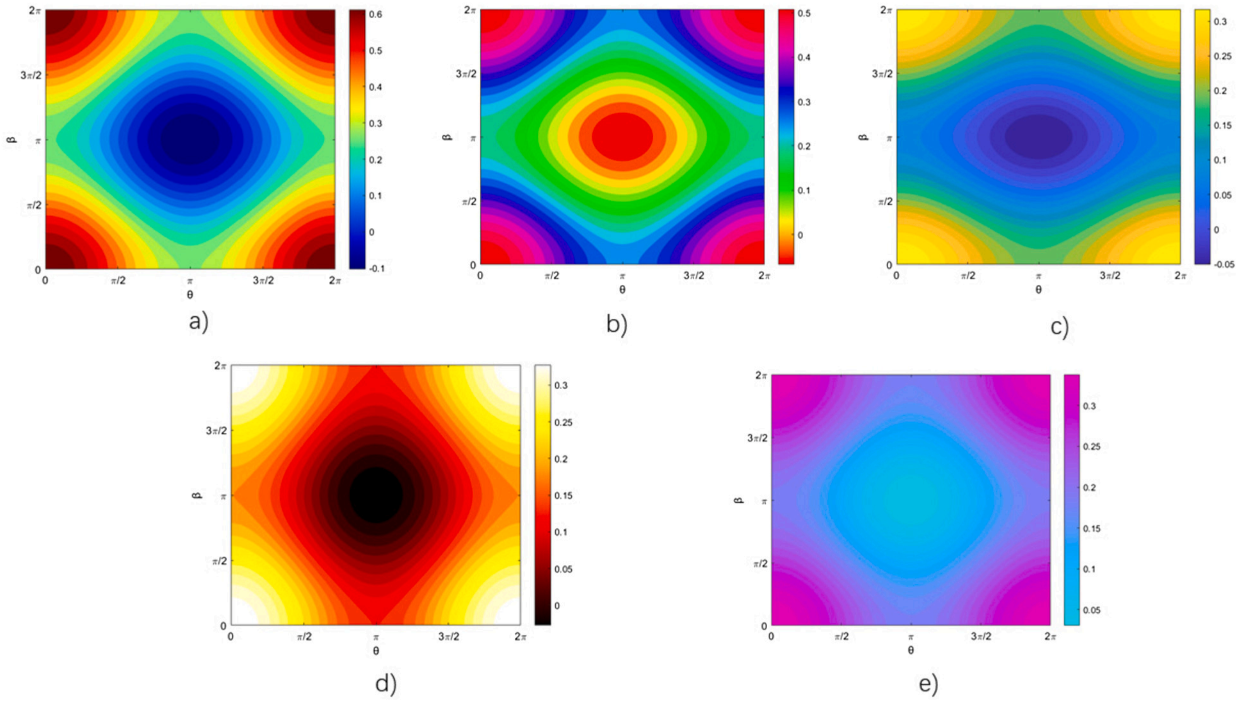

Fig. 8. Probability changes for alternatives based on changes in θ and β.

In order to more intuitively represent probability scores and their interference effects, a method was used to convert the 3D data model into a 2D flat representation. This transformation process leverages specific data visualization techniques to effectively present patterns and trends within the data. In Fig. 8, the results depict the variations in probability scores under different parameter settings, illustrating the intricate interactions among these parameters. Through 2D visualization, key data features are accentuated, offering fresh perspectives for in-depth analysis.

Based on these visualizations, a number of important conclusions can be drawn. These conclusions are mainly concerned with the core change trend of the data model and the influence of the interaction between different parameters on the results. In this way, 2D data visualization not only enhances the understanding of data patterns, but also provides an important basis for analysis of key factors in the decision-making process. The conclusion is as follows:

1) The probability scores of the alternatives exhibit symmetry around (π,π), making it adequate to analyze interference effects within the range $[0,\pi]\times[0,\pi]$. Consequently, when taking interference effects into account, it is only necessary to analyze the data on one side of the symmetry axis to obtain a comprehensive understanding of interference effects. This streamlined approach simplifies the analysis process.
2) The relationship between the probability scores of the alternatives and the phase angles θ and β is non-linear. The result of calculation shows the fluctuation, which can better reflect the interference effect. The nonlinear relationship means that there is not a simple direct proportional relationship between the probability score and the phase change, but a more complex interaction, resulting in a volatility change in the probability score, which more truly reflects the complexity of the interference effect.
3) Within a specified phase range $[0,\pi]\times[0,\pi]$, an increase in phase leads to a more pronounced negative interference effect between any two decision groups. Consequently, the probability score decreases, resulting in a diminished outcome. This observation signifies that, under specific conditions, phase escalation amplifies the adverse interference effect between decision groups, consequently influencing the final decision outcome and resulting in a reduction in the probability score, indicative of a lower likelihood of the outcome. This discovery holds substantial implications for comprehending and managing interference effects in decision-making processes.
4) Within the considered scope, the point $\left[\frac{\pi}{2},\frac{\pi}{2}\right]$ represents a pivotal boundary. At this threshold, decision-making groups operate independently without mutual interference. This implies that, at this juncture, the actions and choices of distinct decision groups remain isolated from one another. However, once deviating from this point, interference effects come into play, influencing the decision-making processes of these diverse groups. As a result of this interference, even decision groups with some similarities witness substantial deviations in their probability scores compared to the ideal point $\left[\frac{\pi}{2},\frac{\pi}{2}\right]$. Understanding and identifying this critical point is paramount for optimizing the decision-making process, especially in scenarios involving the assessment and management of interactions among multiple decision groups.

Table 3
Comparison different methods.

Methods	Ranking indices	Final ranking orders
Classic-TOPSIS [46]	$C_1^* = 0.9027, C_2^* = 0.7791,$ $C_3^* = 0.4817, C_4^* = 0.3126,$ $C_5^* = 0.5275.$	$a_1 \succ a_2 \succ a_5 \succ a_3 \succ a_4$
An extended TODIM [47]	$\phi(x_1) = -24.6199, \phi(x_2) = -10.4131,$ $\phi(x_3) = 8.4470, \phi(x_4) = 15.6519,$ $\phi(x_5) = 10.9341.$	$a_4 \succ a_5 \succ a_3 \succ a_2 \succ a_1$
Quantum framework of [19]	$p(a_1) = 0.3873, p(a_2) = 0.2296,$ $p(a_3) = 0.0544, p(a_4) = 0.0967,$ $p(a_5) = 0.2321.$	$a_1 \succ a_5 \succ a_2 \succ a_4 \succ a_3$
The proposed method	$p(a_1) = 0.4369, p(a_2) = 0.3938$ $p(a_3) = 0.2020, p(a_4) = 0.2769$ $p(a_5) = 0.3647.$	$a_1 \succ a_2 \succ a_5 \succ a_4 \succ a_3$

5) In the range of $\left[\frac{\pi}{2}, \frac{\pi}{2}\right] \times \left[\frac{\pi}{2}, \frac{\pi}{2}\right]$, both factors and produce a positive interference effect, resulting in a disjunctive outcome. This implies that when these factors interact, they favor the selection of one factor over the other in the decision-making process. Conversely, within a different specific range $\left[\frac{\pi}{2}, \pi\right] \times \left[\frac{\pi}{2}, \pi\right]$, these factors yield negative interference effects, leading to conjunctive outcomes. In such cases, the decision tends to consider both factors simultaneously rather than favoring one over the other. In additional ranges, interactions between factors may produce mixed results, encompassing both disjunctive and conjunctive elements or more intricate decision-making patterns.

The current study does not delve into the determination of specific interference values but rather focuses on the overall change in phase angles from 0 to 2π. This approach examines the interference effects among decision-makers based on overall changes, thereby deriving the probability value changes for each alternative solution. Therefore, the future research will aim to quantify the specific phase angles based on existing objective outcomes. This will allow for the calculation of precise interference values, better reflecting the interference effects between decision groups.

5.4. Comparative analysis

To validate the effectiveness of the method presented in this paper, a comparative analysis was conducted with several existing decision-making models. These comparison models could be directly implemented or indirectly adjusted to address the case problem discussed in this paper. The final ranking results derived from different methods are illustrated in the Table 3.

Compared with traditional decision-making models, the ranking of alternative derived from the method proposed in this paper significantly differs. For example, the optimal alternative using the Classic-TOPSIS [46] method is a_1, the TODIM [47] method identifies a_4 as the best alternative, while the results from the model presented in this paper indicate that a_1 is the optimal alternative. This discrepancy arises because the interference effects between decision groups were not considered in the traditional models, leading to subjective influences on the ranking of solutions. By contrast, when comparing with quantum models [19], the optimal alternative remains a_1, but the rankings of other alternatives differ. This is because the proposed method considers the interactions between attributes and evidence fusion, and determines attribute weights through grey relational analysis, resulting in outcomes for each alternative that are more in line with real-world scenarios.

6. Conclusion

Considering the complexity, dynamism, and uncertainty inherent in meteorological disasters, this paper proposes a quantum group decision-making model for meteorological disasters, grounded in evidence theory and the Choquet integral. To address the interplay among attributes, the model integrates information through the 2-additive Choquet integral and evidence theory for effective weight assignment. Furthermore, it employs grey relational analysis to determine the weights of decision-makers based on the correlations among decision groups. Ultimately, the model harnesses the quantum framework to aggregate alternatives information, utilizing quantum probability to prioritize rescue operations optimally.

While, assuming that there are n decision makers, there will be $m = \frac{n(n-1)}{2}$ interference terms when calculating the quantum probability of the scheme. With the increase of the number of decision makers, these interference terms will increase in the form of power series, resulting in an extremely large number of interference terms when n exceeds a specific threshold. In statistics, when $m > 30$, the interference term is a large sample, that is, the input item (number of decision-makers) n is greater than 8, the complexity of calculation using the existing model will increase significantly and become extremely difficult. Therefore, we could solve this problem with a machine learning approach, which has significant advantages in processing large data sets, providing fast computation and low time consumption. Our future research will focus on exploring how machine learning techniques can be effectively used to improve quantum group decision models.

CRediT authorship contribution statement

Shuli Yan: Writing – original draft, Supervision, Methodology. **Yizhao Xu:** Writing – review & editing, Visualization, Data curation. **Zaiwu Gong:** Supervision, Methodology, Conceptualization. **Enrique Herrera-Viedma:** Supervision, Conceptualization.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, and we do not have any possible conflicts of interest.

Data availability

The authors do not have permission to share data.

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References

[1] A. Vafadarnikjoo, K. Chalvatzis, T. Botelho, D. Bamford, A stratified decision-making model for long-term planning: application in flood risk management in Scotland, Omega 116 (2023) 102803, https://doi.org/10.1016/j.omega.2022.102803.
[2] J. Zheng, Y.-M. Wang, K. Zhang, J.-Q. Gao, L.-H. Yang, A heterogeneous multi-attribute case retrieval method for emergency decision making based on bidirectional projection and todim, Expert Syst. Appl. 203 (2022) 117382, https://doi.org/10.1016/j.eswa.2022.117382.
[3] L. Rajput, I. Beg, S. Kumar, Spherical fuzzy analytic hierarchy process and linear assignment model based mcgdm method with its application in ranking of states for their business climate, Expert Syst. Appl. 238 (2024) 122247, https://doi.org/10.1016/j.eswa.2023.122247.
[4] P. Liu, M. Shen, Y. Geng, Risk assessment based on failure mode and effects analysis (fmea) and waspas methods under probabilistic double hierarchy linguistic term sets, Comput. Ind. Eng. 186 (2023) 109758, https://doi.org/10.1016/j.cie.2023.109758.
[5] Y. Li, E. Herrera-Viedma, G. Kou, J.A. Morente-Molinera, Z-number-valued rule-based decision trees, Inf. Sci. 643 (2023) 119252, https://doi.org/10.1016/j.ins.2023.119252.
[6] W. Zayat, H.S. Kilic, A.S. Yalcin, S. Zaim, D. Delen, Application of madm methods in industry 4.0: a literature review, Comput. Ind. Eng. 177 (2023) 109075, https://doi.org/10.1016/j.cie.2023.109075.
[7] C. Sáenz-Royo, F. Chiclana, E. Herrera-Viedma, Ordering vs. AHP. Does the intensity used in the decision support techniques compensate?, Expert Syst. Appl. 238 (2024) 121922, https://doi.org/10.1016/j.eswa.2023.121922.
[8] S. S.K., A. S., An active learning method with entropy weighting subspace clustering for remote sensing image retrieval, Appl. Soft Comput. 125 (2022) 109107, https://doi.org/10.1016/j.asoc.2022.109107.
[9] A. Gaeta, V. Loia, F. Orciuoli, An explainable prediction method based on fuzzy rough sets, topsis and hexagons of opposition: applications to the analysis of information disorder, Inf. Sci. 659 (2024) 120050, https://doi.org/10.1016/j.ins.2023.120050.
[10] D. Yu, L. Sheng, Z. Xu, Analysis of evolutionary process in intuitionistic fuzzy set theory: a dynamic perspective, Inf. Sci. 601 (2022) 175–188, https://doi.org/10.1016/j.ins.2022.04.019.
[11] R. Fang, H. Liao, A 2-additive Choquet integral-based multi-criterion decision-making method with complex linguistic information in drug value assessment, Appl. Soft Comput. 152 (2024) 111198, https://doi.org/10.1016/j.asoc.2023.111198.
[12] G.-F. Yu, D.-F. Li, A novel intuitionistic fuzzy goal programming method for heterogeneous madm with application to regional green manufacturing level evaluation under multi-source information, Comput. Ind. Eng. 174 (2022) 108796, https://doi.org/10.1016/j.cie.2022.108796.
[13] D. Zhang, R. Mesiar, E. Pap, Choquet type integrals for single-valued functions with respect to set-functions and set-multifunctions, Inf. Sci. 630 (2023) 252–270, https://doi.org/10.1016/j.ins.2023.02.038.
[14] M. Divsalar, M. Ahmadi, E. Ebrahimi, A. Ishizaka, A probabilistic hesitant fuzzy Choquet integral-based todim method for multi-attribute group decision-making, Expert Syst. Appl. 191 (2022) 116266, https://doi.org/10.1016/j.eswa.2021.116266.
[15] P. Kakati, S.G. Quek, G. Selvachandran, T. Senapati, G. Chen, Analysis and application of rectified complex t-spherical fuzzy Dombi-Choquet integral operators for diabetic retinopathy detection through fundus images, Expert Syst. Appl. 243 (2024) 122724, https://doi.org/10.1016/j.eswa.2023.122724.
[16] X. Zhang, J. Shang, J. Wang, Multi-granulation fuzzy rough sets based on overlap functions with a new approach to magdm, Inf. Sci. 622 (2023) 536–559, https://doi.org/10.1016/j.ins.2022.11.146.
[17] Y. Liu, X. Liu, J. Jiang, S. Han, A trust-based quantum probabilistic linguistic multi-criteria group decision making model considering interference effect, Comput. Ind. Eng. 184 (2023) 109503, https://doi.org/10.1016/j.cie.2023.109503.
[18] L. She, S. Han, X. Liu, Application of quantum-like Bayesian network and belief entropy for interference effect in multi-attribute decision making problem, Comput. Ind. Eng. 157 (2021) 107307, https://doi.org/10.1016/j.cie.2021.107307.
[19] Q. Wu, X. Liu, J. Qin, W. Wang, L. Zhou, A linguistic distribution behavioral multi-criteria group decision making model integrating extended generalized todim and quantum decision theory, Appl. Soft Comput. 98 (2021) 106757, https://doi.org/10.1016/j.asoc.2020.106757.
[20] I. M.S., A.K. Cherukuri, Decision-making in cognitive paradoxes with contextuality and quantum formalism, Appl. Soft Comput. 95 (2020) 106521, https://doi.org/10.1016/j.asoc.2020.106521.
[21] C. Moreira, P. Tiwari, H.M. Pandey, P. Bruza, A. Wichert, Quantum-like influence diagrams for decision-making, Neural Netw. 132 (2020) 190–210, https://doi.org/10.1016/j.neunet.2020.07.009.

[22] P. Du, B. Zhou, M. Yang, Carbon emission reduction contribution analysis of electricity enterprises in urban green development: a quantum spherical fuzzy sets-based decision framework, Technol. Forecast. Soc. Change 200 (2024) 123181, https://doi.org/10.1016/j.techfore.2023.123181.

[23] G. Kou, D. Pamucar, H. Dinçer, M. Deveci, S. Yüksel, M. Umar, Perception and expression-based dual expert decision-making approach to information sciences with integrated quantum fuzzy modelling for renewable energy project selection, Inf. Sci. 658 (2024) 120073, https://doi.org/10.1016/j.ins.2023.120073.

[24] R. Harikrishnakumar, S. Nannapaneni, Forecasting bike sharing demand using quantum Bayesian network, Expert Syst. Appl. 221 (2023) 119749, https://doi.org/10.1016/j.eswa.2023.119749.

[25] S.E. Borujeni, S. Nannapaneni, N.H. Nguyen, E.C. Behrman, J.E. Steck, Quantum circuit representation of Bayesian networks, Expert Syst. Appl. 176 (2021) 114768, https://doi.org/10.1016/j.eswa.2021.114768.

[26] A.P. Dempster, Upper and Lower Probabilities Induced by a Multivalued Mapping, Classic Works of the Dempster-Shafer Theory of Belief Functions, vol. 219, Springer Berlin Heidelberg, 2008, pp. 57–72.

[27] Q. Xue, F. Xiao, An ambiguity-measure-based complex belief entropy in complex evidence theory, Inf. Sci. 651 (2023) 119726, https://doi.org/10.1016/j.ins.2023.119726.

[28] C. Qiang, Y. Deng, A new correlation coefficient of mass function in evidence theory and its application in fault diagnosis, Appl. Intell. 52 (2022) 7832–7842, https://doi.org/10.1007/s10489-021-02797-2.

[29] L. Xiong, X. Su, H. Qian, Conflicting evidence combination from the perspective of networks, Inf. Sci. 580 (2021) 408–418, https://doi.org/10.1016/j.ins.2021.08.088.

[30] X. Su, S. Mahadevan, P. Xu, Y. Deng, Dependence assessment in human reliability analysis using evidence theory and ahp, Risk Anal. 35 (2015) 1296–1316, https://doi.org/10.1111/risa.12347.

[31] H. Zhang, Feature selection for single cell rna sequencing data based on a noise-robust fuzzy relation and fuzzy evidence theory, Appl. Soft Comput. 148 (2023) 110940, https://doi.org/10.1016/j.asoc.2023.110940.

[32] Z. Li, Q. Zhang, S. Liu, Y. Peng, L. Li, Information fusion and attribute reduction for multi-source incomplete mixed data via conditional information entropy and d-s evidence theory, Appl. Soft Comput. 151 (2024) 111149, https://doi.org/10.1016/j.asoc.2023.111149.

[33] K. Wu, F. Xiao, A novel quantum belief entropy for uncertainty measure in complex evidence theory, Inf. Sci. 652 (2024) 119744, https://doi.org/10.1016/j.ins.2023.119744.

[34] A. Mondal, B.K. Giri, S.K. Roy, An integrated sustainable bio-fuel and bio-energy supply chain: a novel approach based on dematel and fuzzy-random robust flexible programming with me measure, Appl. Energy 343 (2023) 121225, https://doi.org/10.1016/j.apenergy.2023.121225.

[35] B.K. Giri, S.K. Roy, M. Deveci, Fuzzy robust flexible programming with me measure for electric sustainable supply chain, Appl. Soft Comput. 145 (2023) 110614, https://doi.org/10.1016/j.asoc.2023.110614.

[36] Q. Wu, X. Liu, L. Zhou, Z. Tao, J. Qin, A quantum framework for modeling interference effects in linguistic distribution multiple criteria group decision making, IEEE Trans. Syst. Man Cybern. Syst. 52 (2022) 3492–3507, https://doi.org/10.1109/TSMC.2021.3072388.

[37] G. Beliakov, S. James, J. Wu, Discrete Fuzzy Measures, vol. 382, Springer International Publishing, 2020.

[38] M. Grabisch, C. Labreuche, A decade of application of the Choquet and Sugeno integrals in multi-criteria decision aid, Ann. Oper. Res. 175 (2010) 247–286, https://doi.org/10.1007/s10479-009-0655-8.

[39] G. Shafer, A Mathematical Theory of Evidence, vol. 42, Princeton University Press, 1976.

[40] C. Moreira, A. Wichert, Quantum-like Bayesian networks for modeling decision making, Front. Psychol. 7 (2016) 11, https://doi.org/10.3389/fpsyg.2016.00011.

[41] J.-L. Chabert, G. Peruginelli, Adelic versions of the Weierstrass approximation theorem, J. Pure Appl. Algebra 222 (3) (2018) 568–584, https://doi.org/10.1016/j.jpaa.2017.04.020.

[42] A. Mondal, S.K. Roy, Behavioural three-way decision making with Fermatean fuzzy Mahalanobis distance: application to the supply chain management problems, Appl. Soft Comput. 151 (2024) 111182, https://doi.org/10.1016/j.asoc.2023.111182.

[43] L. Shaw, S.K. Das, S.K. Roy, Location-allocation problem for resource distribution under uncertainty in disaster relief operations, Socio-Econ. Plan. Sci. 82 (2022) 101232, https://doi.org/10.1016/j.seps.2022.101232.

[44] K. Debnath, S.K. Roy, Power partitioned neutral aggregation operators for t-spherical fuzzy sets: an application to h2 refuelling site selection, Expert Syst. Appl. 216 (2023) 119470, https://doi.org/10.1016/j.eswa.2022.119470.

[45] B.K. Giri, S.K. Roy, M. Deveci, Projection based regret theory on three-way decision model in probabilistic interval-valued q-rung orthopair hesitant fuzzy set and its application to medicine company, Artif. Intell. Rev. 56 (2023) 3617–3649, https://doi.org/10.1007/s10462-023-10611-3.

[46] R. Susmaga, I. Szczęch, D. Brzezinski, Towards explainable topsis: visual insights into the effects of weights and aggregations on rankings, Appl. Soft Comput. 153 (2024) 111279, https://doi.org/10.1016/j.asoc.2024.111279.

[47] B. Llamazares, An analysis of the generalized todim method, Eur. J. Oper. Res. 269 (3) (2018) 1041–1049, https://doi.org/10.1016/j.ejor.2018.02.054.